



Process abnormality identification by fuzzy logic rules and expert estimated thresholds derived certainty factor

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ARTICLE INFO

Keywords:

Abnormality identification
Rule-based expert system
Certainty factor
Imprecise data
Hydrometallurgy thickening process

ABSTRACT

Abnormality identification plays a significant role in keeping a process healthy and safe. The rule-based expert system (RB-ES) is an efficient approach in this field. However, the relationship between propositions is fuzzy and uncertain when the process knowledge is summarized into rules. Moreover, the process data utilized as observed evidence to activate the uncertain rules face the imprecise measurement problem. The imprecise measurement problem comes from noises, heavy dust, grease pollution, and other circumstance factors that influence sensors. To address the above issues, an abnormality identification framework based on fuzzy logic rules is proposed in this work. Two kinds of uncertain rules with fuzzy linguistic information and expert estimated threshold in premises are specially considered. First, a deviation estimation method based on probability distribution is proposed to extend the imprecise data into the interval data to solve the uncertain problem caused by imprecise measurement. Second, the fuzzy piecewise membership function is utilized to calculate the certainty factor (CF) of the premise in rules, which describes the occurrence risk of an event or abnormality. Third, the inference steps based on the CF expert system with uncertain rules and imprecise activated variables are presented to identify the abnormality in the complex industrial processes. The proposed method is applied to the thickening process in hydrometallurgy and performs better than the traditional rule-based method and the PLS method on the actual process data. The accuracy of the proposed method still maintains higher when more noises are added, generally 8%–10% higher than the other two methods. The proposed method also shows advantages in identifying multi-abnormality and reducing model establishing workload.

1. Introduction

In most industrial processes, the occurrence of abnormality is inevitable because of the interaction of numerous units. An outbreak of abnormality may result in financial losses or even injuries. Abnormality identification in time and corresponding control actions can reduce energy consumption, keep the whole process's stability, and improve the plant's economic benefit. Therefore, abnormality identification plays a significant role in keeping the process healthy. To most industrial processes, abnormality identification is mainly implemented by manual operation. On the one hand, it demands workers' superb skills and causes too much workload. On the other hand, the inference process may contain too much subjective information. It is an independent event to develop new identification methods [1].

Most of the industrial processes have complex procedures. For example, there are five or six sub-processes for the gold hydrometallurgy

under extensive production, but the mechanism of each subprocess is unclear yet. As a result, the mechanism model is hard to keep accurate, and the mechanism-based abnormality identification methods are difficult to be applied [2,3]. The data-driven methods, such as principal component analysis (PCA) and partial least square (PLS), are effective ways to deal with fault detection and abnormality identification [4,5]. However, these methods request the process data precise and believable, which is hard to be satisfied in most actual situations. The imprecision of data results from many factors. For example, noise disturbance may affect the measurement of sensors, or the sensors may drift when they are lack maintenance. A lot of works have been explored by the researchers on data imprecision. The dynamic response of a nonlinear system is investigated, and the uncertain interval analysis is utilized to overcome the data imprecision by Liang [6]. Zhu et al. [7] studied the imprecise DEA model in multiple objective programming. A scale transformation

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<https://doi.org/10.1016/j.chemolab.2020.104232>

Received 17 August 2020; Received in revised form 23 December 2020; Accepted 24 December 2020

Available online 31 December 2020

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technique was taken to overcome the problem of data imprecision. Liu et al. [8] proposed a classification method based on fuzzy logic for imprecise data and received satisfactory performance to the objects easily misclassified between classes. Numerical examples were provided by Lev V. to demonstrate that imprecise interval information can improve the robustness performance of the SVM algorithm [9]. Petridis et al. [10] used interval numbers to predict annual sugar production based on populations of measurements and to improve the prediction accuracy. Besides, Tarek et al. [11] extended conventional PCA to interval numbers and employed the method for process monitoring, which yielded more robust results. Generally speaking, two ways are used to solve the problems caused by imprecise data. The first is to transform the imprecise data into exact data, and the second is to denote the imprecise data in a fuzzy format.

The expert system (ES) is one of the most active and extensive artificial intelligence fields. The ESs emphasize the significance of knowledge and experience. Simultaneously, the smart system and intelligent model become the tendency in the abnormality identification field [12–14]. Among the applications of the ESs, the rule-based ES [15,16] and the case-based ES [17] are the most commonly employed methods. But the case-based ES also has a request for data precision. The RB-ES can explain the causal relationship between events and express process knowledge adapting with human beings' habits. In the meantime, it has been proved efficient in fuzzy and uncertain reasoning [18,19]. Yang et al. [20] extended the fuzzy rule with belief function theory and proposed a new knowledge representation scheme in a belief rule structure [19,21]. However, the method extracts rules from a large amount of data and pays little attention to the rules expressed by linguistic information [22–24]. Even though the linguistic description information is often denoted in the fuzzy form, like 'very high', 'not too much', and so on, it contains much more information than data information and should be drawn more attention.

From the above analysis, two issues should be considered when uncertain problems exist. The first issue is to deal with uncertainties, and the other issue is to represent and measure the uncertain degree. Previous researches on how to measure uncertain degree were carried out from the perspective of probability [25]. The probability can describe the occurrence degree of an event or an abnormality. Nevertheless, the classical probability theory is hardly applied because the prior probabilities or the condition probabilities of events are hard to be obtained in many actual situations. To address this problem, some improvements to classical probability theory were proposed. The D-S evidence theory [26] was proposed to solve the problem of assigning a probability to events 'we don't know whether they will happen or not'. However, how to assign *bpa* (a set of event probabilities in D-S theory) to each evidence and the conflicts in the combination of evidences still confuse the researchers. By introducing two parameters, the subjective Bayesian theory [27] did well in uncertain inference. However, the value range of the two parameters is too large to put into applications. The CF method [28,29] could well substitute the probability, and its good performances were verified in a medical diagnosis system. Compared with the other improvement methods, the CF method can get an accurate conclusion with less computation and more accuracy. However, the assignment for the occurrence degree of an event in the conventional CF method is subjective, which limits the application of this method. Moreover, the assignment is easy to be affected when the industrial processes face imprecise measurement problems.

In this work, an abnormality identification framework based on fuzzy logic rules is proposed to identify the industrial process's abnormality. First, a deviation estimation method based on probability distribution is proposed to extend the imprecise data into the interval data. Then, the fuzzy piecewise membership functions are utilized to calculate CFs of premises in rules. The proposed method is finally applied to the thickening process in hydrometallurgy, which demonstrates its effectiveness and feasibility. The novelty of the proposed method is summarized as follows.

- (1) A fuzzy RB-ES is proposed to identify abnormality, in which the rule premises contain linguistic information or expert estimated thresholds while the activation evidence is imprecise data.
- (2) The imprecise data is extended into the interval data by a probability distribution, which solves the uncertain problem caused by imprecise measurement.
- (3) The CFs of premises in rules are calculated by fuzzy piecewise membership functions, which overcomes the actual application shortcomings of traditional probability.

Briefly, this paper is organized as follows. Section 2 introduces the definition of the CF method and its computing equations. In Section 3, the transformation of imprecise data to interval data and the assignment of CF for initial evidences are introduced. Then, the steps of abnormality identification for the industrial processes with fuzzy rules and imprecise activated variables are introduced. In Section 4, some simulations are presented to show the effectiveness of the proposed method.

2. CF method

Previous researches on how to measure the uncertain degree are carried out from the perspective of probability, and the applications of classical probability all have a condition that the prior probabilities are known beforehand. However, this condition is hard to be satisfied in many actual applications. To address this problem, the CF method [29] is proposed and has been widely employed in uncertain rule-based inference because of the simple operation and less assumption. Compared with probability theory, the CF method describes the uncertainty by CF and avoids using the prior probability or the conditional probability. Two subjective parameters are adopted in the definition of CF, which are the occurrence possibility of an event and the occurrence possibility of the event when an event is observed, respectively. To distinguish with the classical probability, the two parameters are denoted as $pos(H)$ and $pos(H|E)$. Then $CF(H|E)$ can be defined as:

$$CF \left(\begin{array}{c|c} & E \\ \hline H & \end{array} \right) = \begin{cases} \frac{pos(H|E) - pos(H)}{1 - pos(H)}, & pos(H|E) > pos(H) \\ 0, & pos(H|E) = pos(H) \\ \frac{pos(H|E) - pos(H)}{pos(H)}, & pos(H|E) < pos(H) \end{cases} \quad (1)$$

where $CF(H|E) \in [-1, 1]$. The physical meaning of CF differs from that of the classical probability. The $CF(H|E)$ is the increment of credibility after E is observed. In classical probability theory, the conditional probability $pro(H|E)$ will never be less than the prior probability $pro(H)$. However, the relationship will not hold when it comes to $pos(H)$ and $pos(H|E)$ [30, 31]. In most application, $CF(H|E)$ can be given directly by experienced workers for convenience.

Through the analysis of the premise E , the inference equations of the CF method are obtained.

If the premise is composed of more than one sub-premise, then two equations are shown in Eqs. (2) and (3).

If $E = E_1 \text{ and } E_2 \cdots \text{ and } E_n$, then $CF(E)$ can be denoted as follow:

$$CF(E) = \min\{CF(E_1), CF(E_2), \dots, CF(E_n)\} \quad (2)$$

If $E = E_1 \text{ or } E_2 \cdots \text{ or } E_n$, then $CF(E)$ can be denoted as:

$$CF(E) = \max\{CF(E_1), CF(E_2), \dots, CF(E_n)\} \quad (3)$$

The CF method is usually applied to the rule-based expert system. Rules set the relationships between propositions, and CFs describe the related degrees. Then, the CF of a proposition H when the evidence e is observed can be represented as Eq. (4):

$$CF(H|e) = CF(H|E) \times \max\{CF(E|e), 0\} \quad (4)$$

where $CF(H|E)$ represents the credibility of a rule and $CF(E|e)$ means the matching degree between E and e . Considering that $CF(E|e)$ may be a negative number which means E doesn't happen. To accommodate operators' habit of accepting the description situated in $[0, 1]$, the maximizing function is employed.

If two independent pieces of evidence e_1 and e_2 both indicate the occurrence of H , denoted as $CF_1(H)$ and $CF_2(H)$ respectively, the final CF of H is calculated as:

$$CF(H) = \begin{cases} CF_1(H) + CF_2(H) - CF_1(H) \times CF_2(H) & CF_1(H) \geq 0 \text{ and } CF_2(H) \geq 0 \\ CF_1(H) + CF_2(H) + CF_1(H) \times CF_2(H) & CF_1(H) < 0 \text{ and } CF_2(H) < 0 \\ \frac{CF_1(H) + CF_2(H)}{1 - \min\{CF_1(H), CF_2(H)\}} & CF_1(H) \times CF_2(H) < 0 \end{cases} \quad (5)$$

3. Methodology

In this section, a deviation estimation method based on probability distribution is proposed to extend the imprecise data into the interval data. Then, fuzzy piecewise membership functions are utilized to calculate CFs of premises in rules, which describe the occurrence degree of a proposition. Finally, the abnormality identification method is introduced in detail.

3.1. Real value estimation of imprecise data

On account of errors and disturbances, some variables are measured imprecisely in most actual cases. In this section, a deviation estimation method is discussed. Even though the deviation of a sensor is marked when the sensor is produced. The real deviation will not always keep the same when the working circumstances change, especially in severe industrial circumstances. When some imprecise data are collected online by sensors, some of them can also be tested offline by precise instruments at the same time. However, the testing by the instrument is time-consuming or money-costing. Thus, only a handful of precise samples are collected by the precise instrument. The actual deviation of the variable is calculated by the precise instrument and the corresponding online measured data by sensors as

$$DEV = Value_{\text{precise}} - Value_{\text{sensor}} \quad (8)$$

The population of DEV can be seen as normally distributed $DEV \sim N(\mu, \sigma^2)$, where μ is the expectation of DEV and σ^2 is the variance.

Our purpose is to get μ by analyzing the samples of the DEV . The t -distribution hypothesis test is utilized to estimate a population's expectation μ by a small number of samples [32].

Suppose the average value of the DEV samples is denoted as $\overline{DEV}$ and the variance of samples is S , let

$$t = \frac{\overline{DEV} - \mu}{S/\sqrt{n}}$$

Then, t follows the t -distribution with the degree of freedom $n - 1$, denoted as

$$\frac{\overline{DEV} - \mu}{S/\sqrt{n}} \sim t_{\alpha}(n-1)$$

Rewrite as:

$$P\left(\frac{\overline{DEV} - \mu}{S/\sqrt{n}} \geq t_{\alpha}(n-1)\right) = \alpha \quad (9)$$

where n represents the sample number. α means the rejection region of

the hypothesis. α usually takes 0.01 or 0.05. The value of μ can be calculated at the boundary condition when the equality of Eq. (9) still holds. $t_{\alpha}(n-1)$ can be obtained from the t -distribution lookup table, denoted as a . Then,

$$\frac{\overline{DEV} - \mu}{S/\sqrt{n}} = a \quad (10)$$

$$\mu = \overline{DEV} - a \cdot S / \sqrt{n} \quad (11)$$

When the deviation expectation is obtained according to Eq. (11), the imprecisely measured data can be extended to the interval format [33, 34]. If the value of online measured data is b and $\mu \geq 0$, then the measured value is extended to the interval $[b, b + \mu]$ according to Eq. (8). Otherwise, if $\mu < 0$, the value is extended to $[b + \mu, b]$. We can also take σ^2 into account and extend the value to a much broader interval. The chi-square distribution hypothesis test can be utilized to estimate the value of σ . For space limitation, it is not introduced in this paper anymore. In our research, only the expectation of the population μ is considered when the measured data is extended.

3.2. CF assignment method for initial evidences based on fuzzy piecewise membership functions

Most of the abnormality identification rules in rule-based methods are extracted from process knowledge. The conclusions of the rules are operating conditions, while the premises of rules are constituted with linguistic information or expert estimated thresholds. These two kinds of rules are first discussed in this section. The rules with indicator variable thresholds in premises, are shown as follow:

$$\text{if } x_1 \oplus \text{threshold}_1, x_2 \oplus \text{threshold}_2, \dots, x_n \oplus \text{threshold}_n, \text{ then } H \text{ (CF)} \quad (12)$$

where $\oplus$ means a relationship between the observed data value and a certain threshold, it can be equalities or inequalities. H is the conclusion indicating the process is normal or abnormal. CF is the certainty factor that describes the occurrence degree of the conclusion.

Sometimes, the rules extracted from process knowledge by experts are of linguistic properties inevitably. For example, an expert may set a rule like 'if the slurry pump's current is very high, the slurry concentration is high'. This kind of uncertain rule is expressed as:

$$\text{if } E(\text{fuzzy linguistic description}) \text{ then } H \text{ (CF)} \quad (13)$$

where E may contain many fuzzy descriptions, such as 'too low', 'too high', 'higher than', and so on. H is the conclusion and CF is the certainty factor.

The distributed control systems make data collection faster and easier. A large amount of data is available when the process is undergoing. Even though the rules in RB-ES contain lots of fuzzy premises, the online data is the evidence to activate the premises of rules. Therefore, the first step of the uncertain inference is to assign CF for each premise based on the observed evidence.

In our research, membership functions are established for every fuzzy premise. For example, if the value of a corresponding variable is observed, a membership function is invoked to calculate the matching degree to a fuzzy premise. CFs present the match degrees. Therefore, a CF assignment method for the initial evidence is proposed.

The fuzzy piecewise membership functions are adopted for each fuzzy premise. For example, membership of observation point-valued x_0 for a 'very high' premise is set as below:

$$CF(\text{very high}|x_0) = \begin{cases} 1 - \exp(j \cdot (x_{\text{mean}} - x_0)) & x_0 \geq x_{\text{mean}} \\ \exp(j \cdot (x_0 - x_{\text{mean}})) - 1 & x_0 < x_{\text{mean}} \end{cases} \quad (14)$$

where j is the parameter representing the steepness of the curve.

For the rules containing explicit thresholds, the numerical range is

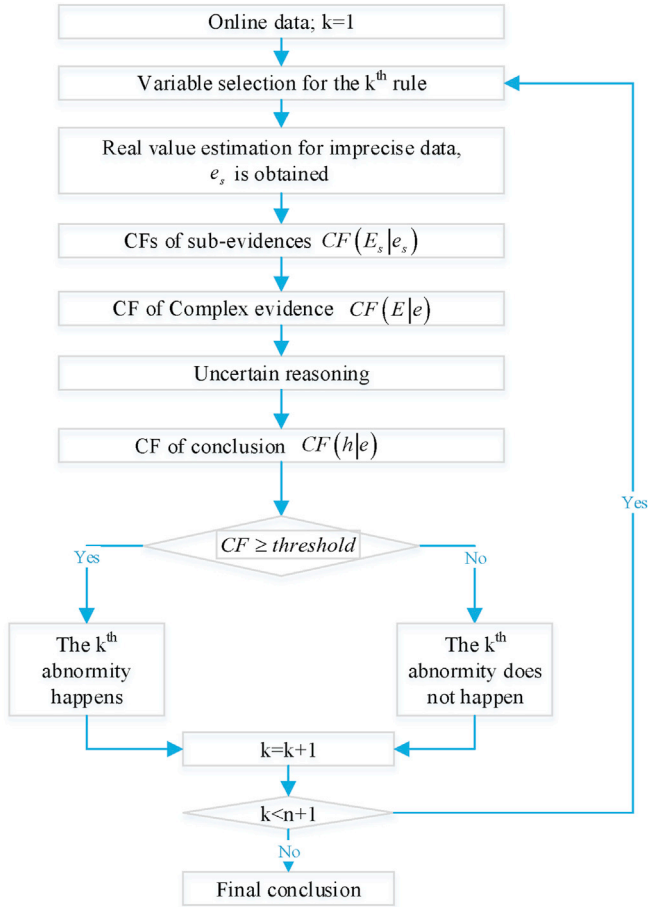


Fig. 1. The flowchart of the rule-based inference with the CF method.

divided into two parts. For example, the membership of observation point-valued x_0 for a 'higher than a certain threshold' premise is set as below:

$$CF(x > threshold|x) = \begin{cases} 1 & x > threshold \\ 0 & otherwise \end{cases} \quad (15)$$

A similar way can be utilized to get the membership function for the 'lower than' premise.

Suppose $x = [a, b]$ is the extended interval observed variable, and it covers r segments of the membership function. The expectation CF for a fuzzy premise is calculated as

$$E\{CF(\text{fuzzy premise}|[a, b])\} = \frac{\sum_{j=1}^r \int_{l_j}^{u_j} g_j(x) dx}{b-a} \quad (16)$$

where $g_j(x)$ is the j^{th} segment of the piecewise membership function. l_j and u_j are the lower and upper limit of the j^{th} segment integral, respectively.

It should be noted that the membership function can only determine the CF of a single piece of evidence. If the premise is comprised of more than one sub-premise or the proposition is reflected by more than one variable, the corresponding equations in Section 2 are employed.

3.3. Abnormity identification steps

In this section, the abnormity identification method for industrial processes with fuzzy rules and imprecise activated data is proposed. In this work, all the premises of the rules contain linguistic information or expert estimated thresholds. And some of the activated data are measured imprecisely. Suppose the expert system contains **Error! Not a valid embedded object.** rules, and each rule associating with a single abnormal working condition. The abnormity identification steps are shown below. And the schematic diagram of the inference steps is shown in Fig. 1.

- 1) $k = 1$;
- 2) Choose corresponding activated variables for the k^{th} rule;
- 3) Calculate the CF for each initial sub-premise in Section 3. If the sub-premise is reflected by more than one activated variable, Eq. (5) is employed to calculate the CF of the sub-premise $CF(E_s|e_s)$.
- 4) Eq. (2) and Eq. (3) are employed to calculate the composite CF of the premise $CF(E|e)$ if the premise is composed of more than one sub-premise. Otherwise, $CF(E|e) = CF(E_s|e_s)$.
- 5) The conclusion $CF(h|e)$ is obtained by Eq. (4).
- 6) Conclude the k^{th} rule. The abnormality happens if the CF value exceeds a preset threshold; otherwise, the abnormality does not happen.
- 7) $k = k+1$, return to step 2), traverse all the rules in the rule base, and calculate their CFs of conclusions.

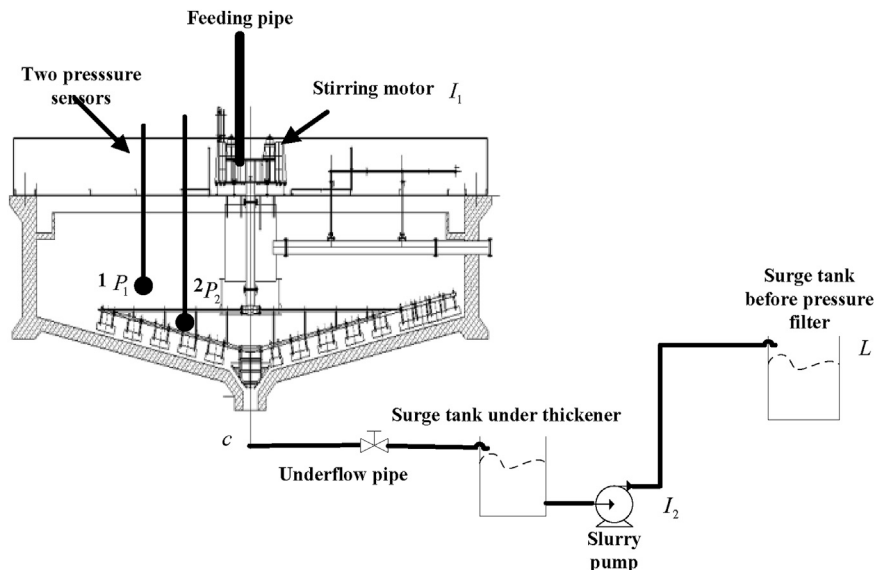


Fig. 2. Brief structure of a thickening process.

Table 1
Measurable variables in the thickening process.

Variable name	Symbol	Average value
Pressure 1	P_1	45 kPa
Pressure 2	P_2	65 kPa
Current of the stirring motor	I_1	12A
Liquid level of the surge tank in front of the pressure filter	L	400 cm
Current of the slurry pump	I_2	26A
Underflow concentration	c	45%

8) Draw a final conclusion for the system.

4. Abnormity identification for the thickening process

In this section, several simulations on the thickening process are designed to demonstrate the effectiveness of the abnormity identification method.

4.1. Introduction of the thickening process

The thickening process [12,35] is widely applied in hydrometallurgical enterprises. The inner mechanism is unclear for us by far because of the severe circumstances. Therefore, abnormity identification and control operations are still handled manually. The structure of the thickening process is shown in Fig. 2.

First of all, the low-concentration slurry is fed into the thickener from the top center feeding pipe. Then, the ore in the slurry precipitates to the bottom because of gravity. The stirring motor drives the rake at the bottom of the thickener to make the slurry precipitate evenly and slowly. Two pressure sensors are employed to estimate the underflow concentration. A large amount of caliche generates on the inner surface of the pressure sensors. If too much deciduous caliche blocks the outlet of the underflow pipe, both the underflow rate and slurry's concentration are influenced. The underflow rate is an important index. On the one hand, the surge under the thickener should not be pumped empty. On the other hand, the surge in front of the pressure filter should not be poured full to overflow. The overflow of the surge tank in front of the pressure filter is a severe production accident. The slurry concentration abnormity, underflow rate abnormity, and overflow of the surge are the three main abnormities in the process.

Six variables are measured in the process, namely the two slurry pressures, the underflow concentration, the current of the stirring motor, the liquid level of the surge in front of the pressure filter, and the current of the slurry pump. The underflow rate Q is immeasurable but significant to identifying the underflow rate abnormity. However, it can be indirectly reflected by the slurry pump's current I_2 . The details of the measurable variables are summarized in Table 1, and their measure points are labeled in the corresponding position in Fig. 2. The severe circumstances of the thickening process make all the variables except the two currents I_1 and I_2 measure imprecisely.

4.2. Model establishment

In the later part of this work, the PLS method and the traditional rule-based method are employed as the contrast methods to identify the industrial processes' abnormity. In this section, the rule bases and the PLS model are established.

Owing to the severe working circumstances, the abnormity identification and healing control both depend on the manual operation by far. The abnormity identification rules for the thickening process are first extracted from process knowledge and working experience. For the traditional rule-based method, the uncertainty problem is neglected. The rules are entirely trusted and will be activated by the collected data if the corresponding variables satisfy the evidences no matter the data is

Table 2
Abnormity identification rules for the thickening process.

Rule number	If-clause	Then-clause	CF of the rule
1	$c > 45\%$	Underflow concentration	0.9
2	$I_1 > 12\text{A}$ or $P_1 > 45\text{kPa}$ or $P_2 > 65\text{kPa}$	abnormity	0.4
3	$I_2(t) < 22\text{A}$	Underflow rate abnormity	0.9
4	$L > 450\text{cm}$	Overflow	1

Table 3
Two causes for the high concentration.

Rule number	If-clause	Then-clause	CF of the rule
1	$c > 45\%$ and Q is low	high concentration resulting from a too low underflow rate	0.9
2	$c > 45\%$ and Q is not low	high concentration resulting from too many slurry feedings	0.9

precise or not. In the proposed method, the CF is attached to each rule to describe the belief degree. The uncertain rules for the main abnormities are shown in Table 2.

The 1st and the 2nd rules in Table 2 are set for the underflow concentration abnormity. If the imprecisely measured concentration is higher than 45%, the system is believed abnormal with the belief degree of 0.9. If the current of the stirring motor I_1 is higher than 12A or the two pressures are higher than 45kPa and 65kPa, respectively, the underflow concentration is abnormal, also with a 0.9 belief degree. The underflow rate Q is unmeasured, and it can only be reflected by the slurry pump's current I_2 . If I_2 is lower than 22A, the underflow rate is abnormal. The 4th rule is to identify the overflow abnormity. If the liquid level is higher than 450 cm, the tank will undoubtedly overflow.

Deep reasons for the underflow concentration abnormity must be traced because the underflow concentration is one of the most important quality indexes. Two reasons account for the abnormity, i.e., too many feedings or too low underflow rate. The reason trace uncertain rules for underflow concentration abnormity are summarized in Table 3.

Two variables are utilized in tracing the abnormity reason, where c is the imprecisely measured underflow concentration and Q is the linguistic description of the underflow rate. Even though the underflow rate is unmeasured, it can be deduced from the value of I_2 . The fuzzy piecewise

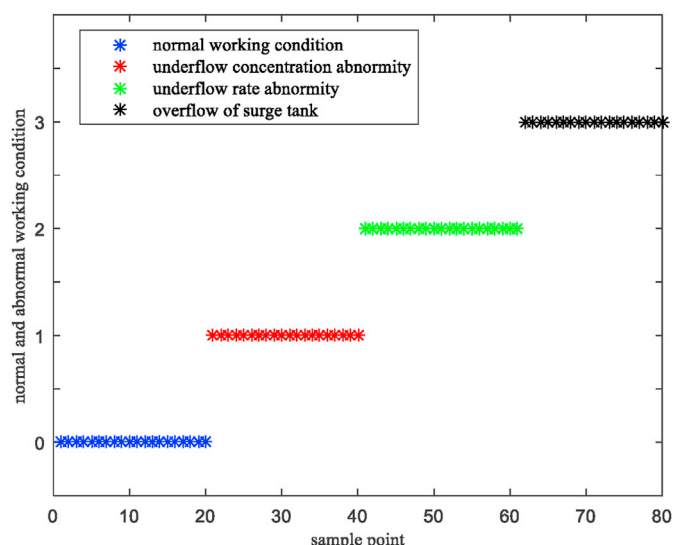


Fig. 3. Identification results of the proposed method.

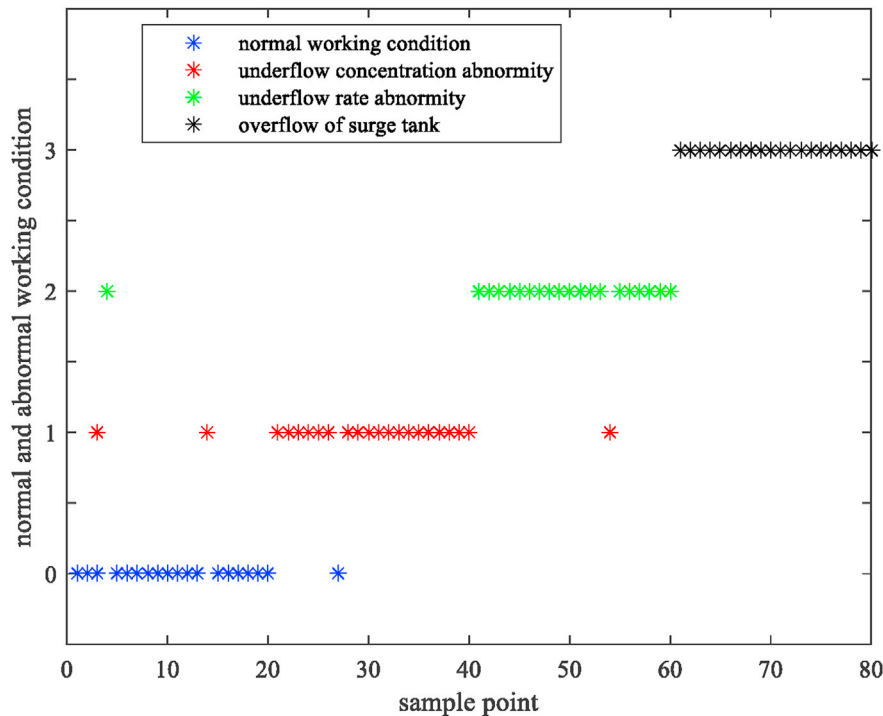


Fig. 4. Identification results of the traditional rule-based method.

membership functions for ‘Q is low’ and ‘Q is not low’ are given in the following section.

It is noted that two kinds of premises exist in the reason trace rules, and the thresholds in Tables 2 and 3 are set according to the manufacturing technique.

4.3. Abnormity identification and analysis

In this section, several cases are designed to demonstrate the value of the proposed method in identifying abnormalities under different amplitude noises and in actual application.

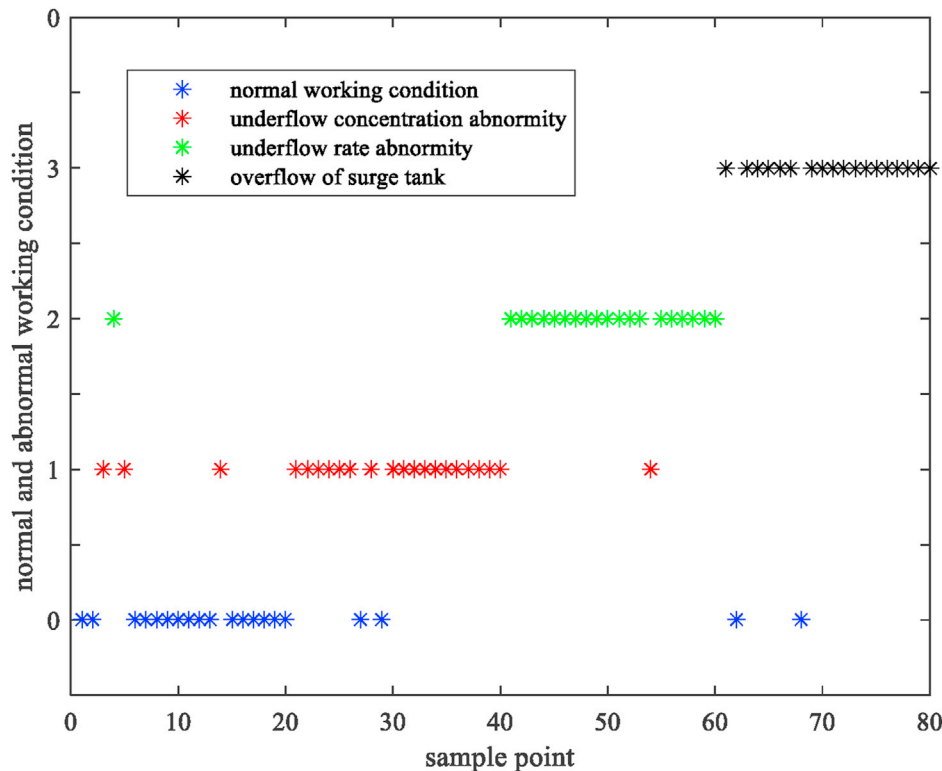


Fig. 5. Identification results of the PLS.

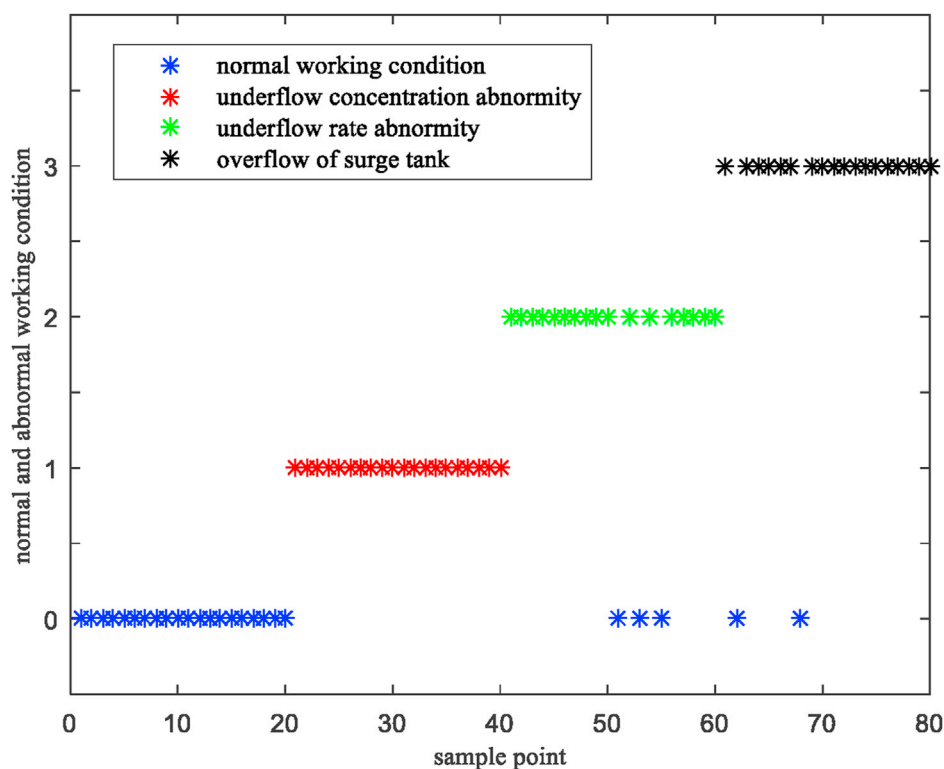


Fig. 6. Identification results of the proposed method with 10% added white noise.

Case #1. The abnormality identification method is firstly applied to the data collected from the actual thickener introduced in Section 4.1. All the data follows the characters of working conditions in Table 2 with small variance, and their operating conditions have been verified in practice. Eighty samples are chosen as the testing data, where the first 20 samples are the normal condition, and the other samples are all abnormal

samples. All the abnormal samples contain a single abnormality. The 21st-40th samples are the high concentration abnormality, which can be identified by the 1st and the 2nd rule. The 3rd rule can identify the 41st-60th samples, which contain a low underflow rate abnormality. The 61st-80th samples are the overflow abnormality of the surge tank in front of the pressure filter. All the samples are firstly identified by three methods, i.e.,

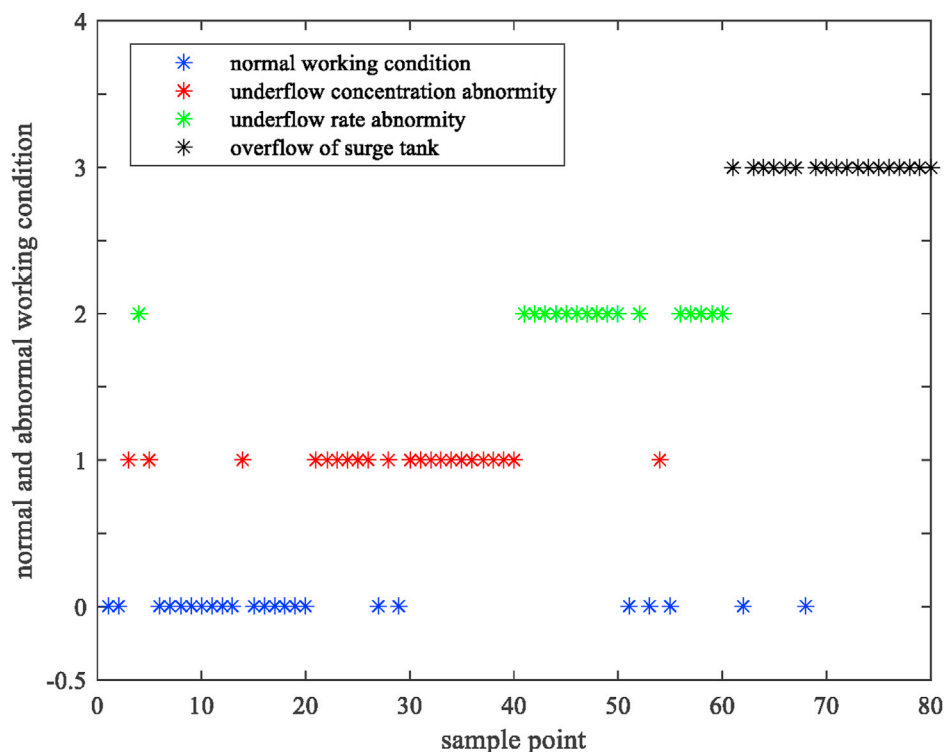


Fig. 7. Identification results of the traditional rule-based method with 10% added white noise.

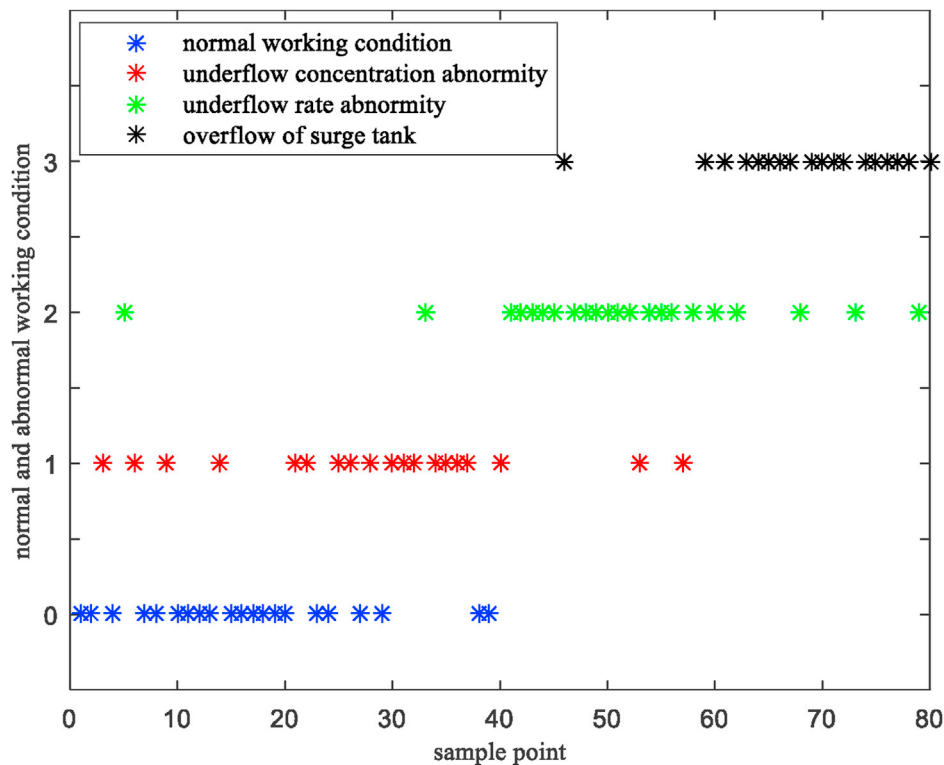


Fig. 8. Identification results of the PLS method with 10% added white noise.

the proposed rule-based method, the traditional rule-based method, and the PLS method. For the proposed method, the CFs of initial evidences are calculated by fuzzy piecewise functions while the CF is given manually in the traditional rule-based method. The identification results by the proposed method are satisfied, as shown in Fig. 3. All the samples are identified correctly. The number 0–3 in the ordinate means the normal working condition, the concentration abnormality, the underflow rate abnormality, and the overflow abnormality. The fuzzy piecewise function can well handle the imprecise measurement problem. The identification results of the traditional rule-based method are shown in Fig. 4. Five samples are identified incorrectly. The PLS method has a data precision request, and nine samples are identified incorrectly because of data imprecision. The identification results are shown in Fig. 5.

Case#2. 10% white noise is added to the six variables of all the 80 samples. The same three methods in Case#1 are employed to identify the abnormalities again. The identified results are shown in Fig. 6, Fig. 7, and Fig. 8, respectively. More noises can affect the identification accuracy of all three methods. In Fig. 6, only five samples are misclassified by the proposed method.

In comparison, 15 samples are misidentified by the traditional rule-based method because some rules are activated incorrectly by the imprecise data. The proposed method has a more robustness property than the traditional rule-based method when more noises are added in the process data. The identification results of the PLS method are shown in Fig. 8, in which 20 samples are misidentified.

To avoid randomness, 4000 accumulated history samples are identified by the three methods again, and each working condition contains

1000 samples. All the modeling samples for the three methods are believed offline test precise samples and the validation samples are imprecise measured. The accuracy of the methods is shown in Table 4.

In comparison, PLS is the typical data-driven method, which has a high request of data precision. It can only identify about 89.3% of samples when the imprecise measurement problem exists or additional noises appear. The traditional rule-based method can identify 92.5% samples correctly while the accuracy of the proposed method raises to 96.5%. The abnormality thresholds for the two rule-based methods are all 0.6. Fuzzy piecewise membership functions are utilized to assign the CFs of initial evidence for the proposed method. The additional noise can reduce the accuracy of all the methods, but the tendency of the three methods maintain the same. The proposed method has a better performance when noises are added or data imprecision exists.

After all the abnormalities are identified, the reason for the underflow concentration abnormality should be traced based on the rules in Table 3. The abnormality reason for another online 400 continuous samples, which have been identified as the underflow concentration abnormality, is traced in the following part. The abnormalities come from a low underflow rate from the beginning to the 100th sample point. The underflow rate rises to a high level from the 101th sampling point. Along with the rise of the underflow rate, the reason changes from low underflow rate to too many slurry feedings.

According to the CF assignment method for the initial evidences in Section 3.2, the piecewise membership function for the linguistic premise ‘the underflow rate Q is low’ is:

$$CF(\text{the underflow rate } Q \text{ is low} | I_2) = \begin{cases} \exp(130 \cdot (26 - I_2)) - 1 & I_2 \geq 26 \\ 1 - \exp(130 \cdot (I_2 - 26)) & I_2 < 26 \end{cases}$$

The piecewise membership function of ‘the underflow rate is not low’ can be easily obtained by symmetric operations to the functions above. After the CFs of initial premises are all assigned, the reason for high concentration abnormality can be obtained by the uncertain inference. All the modeling samples for the two methods are offline test samples, and the online 400 samples are imprecise measured. The high concentration

Table 4
Accuracy of three methods.

Method	Accuracy	
	Case#1	Case#2
PLS	89.3%	83.1%
Traditional rule-based method	92.5%	84.7%
The proposed method	96.5%	93.2%

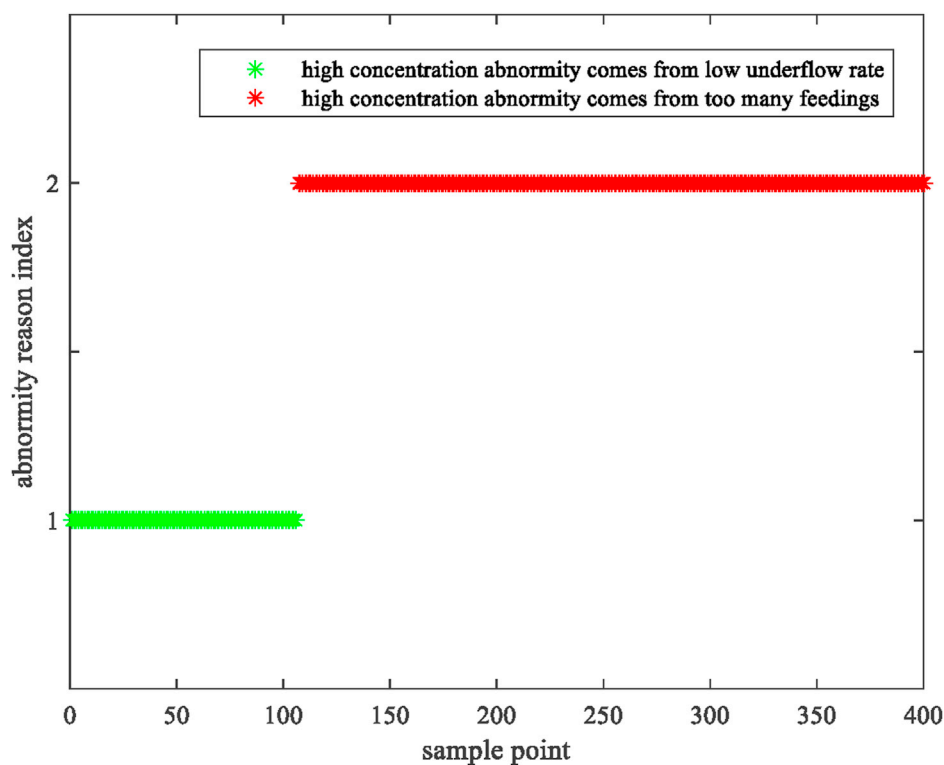


Fig. 9. Reason trace results of the proposed method.

reasons are correctly tracked by the proposed method, as shown in Fig. 9. The traditional rule-based method and the PLS method are utilized to trace the abnormality reason as contrast methods, and the reason trace results are both unsatisfactory, as shown in Fig. 10 and Fig. 11.

In the above analysis, all the samples are running under conditions with a single abnormality. However, in actual applications, there may be

more than one abnormality at the same time. For example, two condition samples are selected from accumulated history data of the same thickener. Details of the two condition samples are shown in Table 5.

According to the production record, both the two samples consist of two abnormalities. The first condition contains high concentration and low underflow rate abnormality, while the second condition contains high

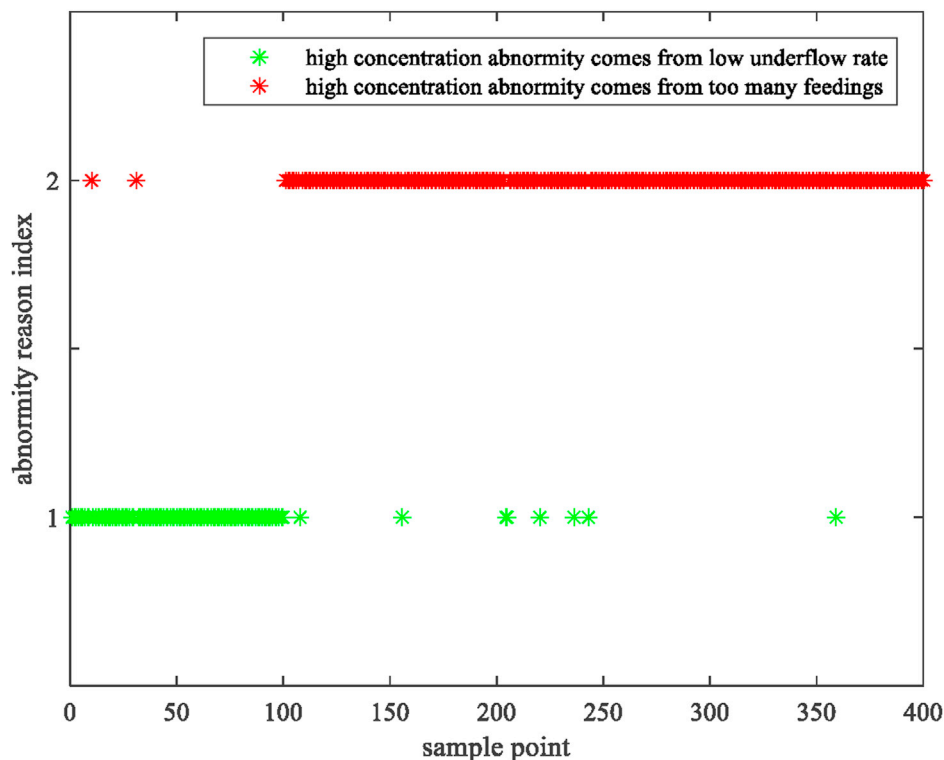


Fig. 10. Reason trace results of the traditional rule-based method.

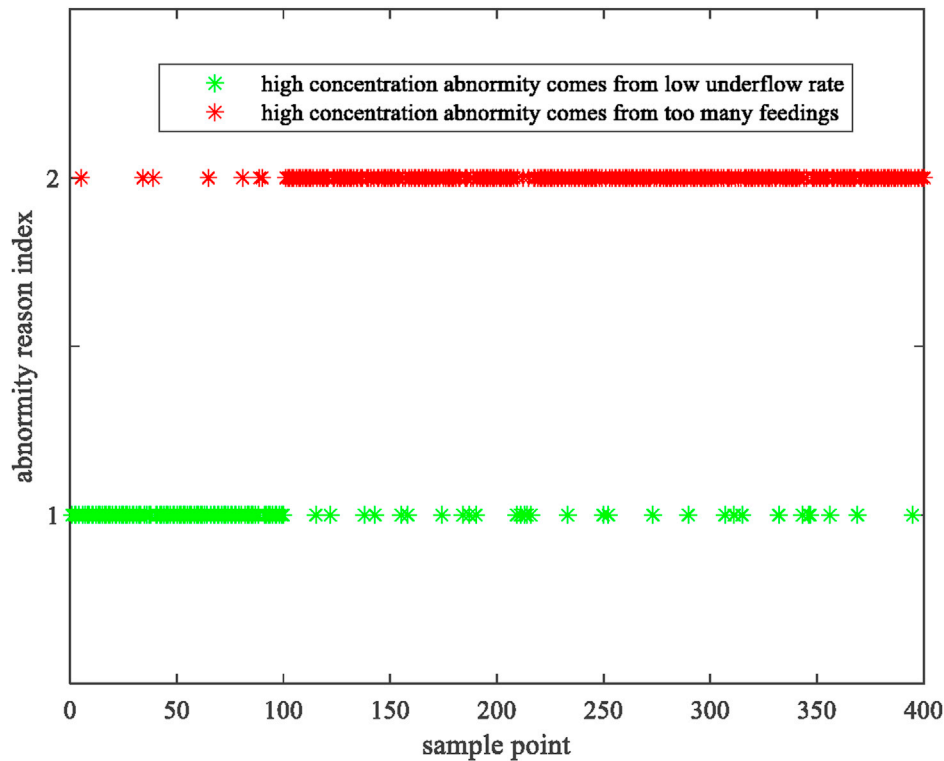


Fig. 11. Reason trace results of PLS.

concentration and overflow abnormality. The variables are all imprecise measured data except the two currents. Take the underflow concentration as an example, the concentration is extended to the interval format, and 31 offline test samples are employed to estimate the deviation population's expectation. The $\overline{DEV}$ and S are separately calculated as 1.358 and 0.509 by chi-squared distribution. Then $\mu_c = 1.2$, cis is extended to [45.7, 46.9] and [49.2, 50.4]. The two pressures and the liquid level of tanks can also be handled in the same way. $\mu_{p1} = -2.3$, $\mu_{p2} = -1.9$, and $\mu_L = -0.6$. All the imprecise data can be extended to interval data finally.

All the CFs of the initial evidences are calculated, and uncertain inference is carried out for both the two conditions. For the 1st condition, the CF of the high concentration abnormality and low underflow rate abnormality both exceed the threshold. In contrast, CFs of other abnormalities are all zeros, as shown in Fig. 12. For the 2nd condition, the high concentration abnormality and the overflow abnormality are both identified. And the CF corresponding to the high concentration is 0.94, which is combined by two rules. The CF of the 1st rule is 0.9, and the CF of the 2nd rule is 0.4, which is shown in Fig. 13.

Both abnormalities in the two conditions are identified by the proposed method without updating the rules. The proposed method can handle the multi-abnormality identification problem without raising the workload. A simulation Case #3 is carried out to illustrate the advantages of the proposed method in actual application.

Case #3. One hundred and twenty pieces of samples are chosen as the testing data, where the first 80 samples are the same as that in Case #1,

Table 5
Details of the two samples in the simulation.

Variable	Condition #1	Condition #2
P_1	42.4 kPa	44.4 kPa
P_2	61.2 kPa	63.7 kPa
I_1	12.1A	12.7A
L	425 cm	451 cm
I_2	17.3 A	27.8A
c	45.7%	49.2%

the 81st-100th samples contain the same abnormality in Condition #1, and the 101th-120th samples contain the abnormalities in Condition #2. All the samples are identified by the proposed method, the PLS model of Case #1 and a corresponding updating PLS model, respectively. The identification result of the proposed method is shown in Fig. 14. Only three multi-abnormality samples are misidentified. However, the PLS model established in Case #1 cannot identify multi-abnormality samples completely. All the 40 multi-abnormality samples are misidentified as single abnormality, as shown in Fig. 15. The Condition #1 and Condition #2 samples are added in the updating PLS model, and the identification result of the updating model is shown in Fig. 16. Even though most of the two kinds of multi-abnormality conditions are identified, the PLS model's total accuracy decreases slightly compared with that in Case #1 because the rise of the class number affects the distinction between classes.

To maintain the PLS model's accuracy, new modeling samples should be added to retrain a new model, which raises the workload of engineers but receives unsatisfactory results. In contrast, the proposed method can handle the rule-based inference method with imprecise activated data and linguistic information. Furthermore, the proposed method can reduce the modeling workload for engineers in actual application.

5. Conclusion

In this work, an abnormality identification framework based on fuzzy uncertain RB-ES is proposed. The relationship between propositions is fuzzy and uncertain when the process knowledge is summarized into rules and the imprecise process data is utilized as observed evidence to activate the rules. The imprecise point data is extended to interval format through a statistical approach, and fuzzy staircase membership functions are utilized to calculate CFs for initial premises in rules. The proposed method can effectively solve the uncertain problem caused by imprecise measurement in the industrial processes. Several simulations are carried out for a thickening process, which have shown the effectiveness of the proposed abnormality identification method. The proposed method can be applied to all the industrial processes facing the imprecise measurement

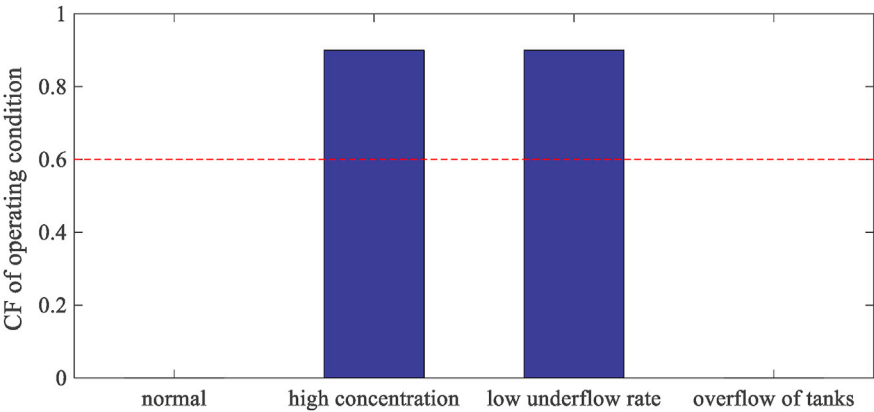


Fig. 12. CFs of the abnormalities for Condition #1.

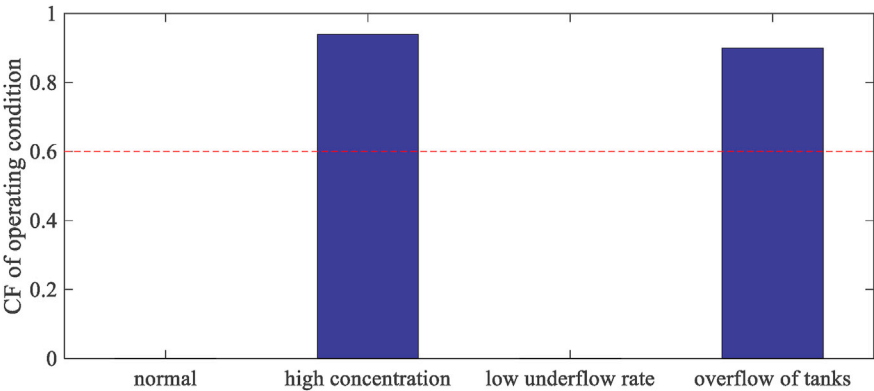


Fig. 13. CFs of the abnormalities for Condition #2.

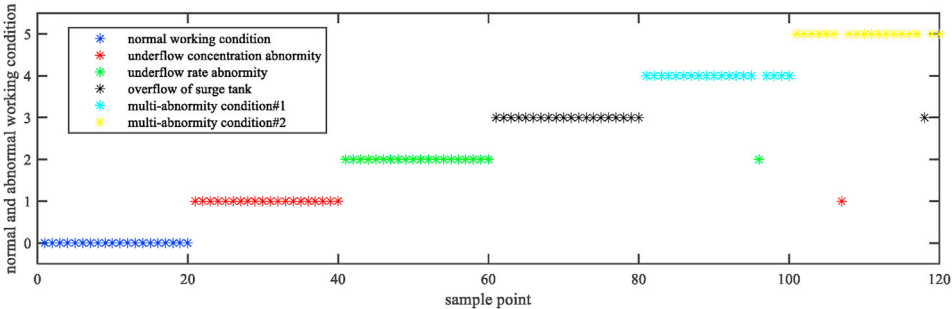


Fig. 14. Identification results of the proposed method for multi-abnormality condition.

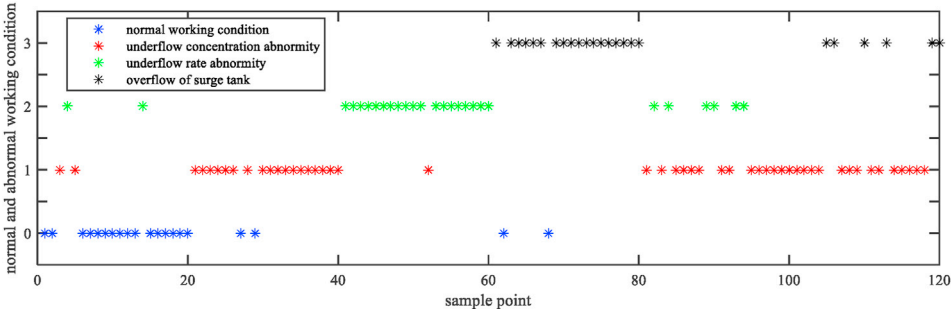


Fig. 15. Identification results of the PLS method without retrained model.

problem in terms of theory. However, from the view of the actual application, the rule-based methods have shortcomings for a too complex

system with hundreds of variables inevitably. The efficiency of the proposed method will be reduced either in the model establishment or

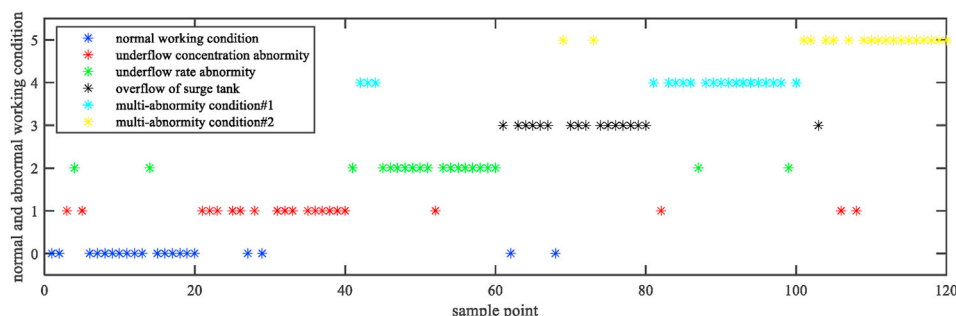


Fig. 16. Identification results of the PLS method with retrained model.

online inference. The belief rule-based inference method can extract rules from process data automatically, which raises the efficiency of the rule-based method. Further research on the combination of the belief rule-based inference method and the proposed linguistic disposal technique in the manuscript will be explored.

CRedit author contribution statement

Yuan jie: programming, simulation and, Writing – original draft, preparation. **Zhang shumei:** simulation and, Methodology. **Wang shu:** Supervision, writing reviewing and editing. **Wang fuli:** Conceptualization, Methodology. **Zhao luping:** writing reviewing and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors gratefully acknowledge the support from the following foundations: the National Key R&D Program of China (2019YFE0105000), the National Natural Science Foundation of China (Nos. 61973057, 61873049, 61503069 and 61903272), the Natural Science Foundation of Tianjin (No. 20JCQNJC01670), and the Open Foundation of State Key Laboratory of Process Automation in Mining & Metallurgy (BGRIMM-KZSKL-2018-09).

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